

Energy-Efficient Transmission Strategy With Dynamic Load for Power Line Communications

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Abstract—To improve energy efficiency (EE) in power line communication (PLC) systems, we proposed a dynamic load-based PLC system model as a new model for EE maximization and an energy-efficient resource allocation strategy optimizing load impedance, transmission power and subchannel allocation as the optimization arguments. Since the load impedance at receivers is influenced by characteristics of a power line channel, optimizing the load impedance is required to maximally induce a received power while considering the channel characteristics. We sought to maximize network EE while satisfying constraints that transmission power of a transmitter cannot be exceeded by its maximum limit and minimum quality of service should be guaranteed. Therefore, we studied a scenario optimizing the three arguments based on orthogonal frequency division multiplexing downlink networks with the non-white Gaussian noise channel in multi-receiver PLC systems. Using nonlinear fractional programming and Lagrange dual method, we provided a tractable solution as an iterative algorithm obtaining the optimal value of the arguments. Simulation results showed that the proposed system is more energy-efficient compared to baseline schemes, and EE is greatly improved by the synergistic effects of the impedance optimization and the subchannel allocation strategy.

Index Terms—Power line communications, energy efficiency, nonconvex optimization, channel impedance, resource allocation.

I. INTRODUCTION

POWER line communications (PLCs) using the power grid as a communication medium have various applications, such as smart-home PLC networking [1], smart distribution grids [2] and automatic metering infrastructure (AMI) [3].

With increasing Internet of Things devices in home-area power line networks, resource allocation to implement energy-efficient information transmission systems with throughput maximization is important. Various optimization

solutions for efficient resource allocation in PLC systems have been studied. In [4], jointly utilizing signal detection and resource allocation techniques have been proposed to enhance energy efficiency (EE) of PLC systems. In [5], the performance improvement of indoor PLC networks under power spectral density limitations with multiple links is achieved by an efficient multiuser-loading algorithm optimizing subcarrier and power allocation. In [6], a bit-loading lookup table resource allocation algorithm is proposed to improve resource utilizing efficiency. In [7], the optimal linear periodically time varying (LPTV)-aware bit loading is studied to save power and improving bit loading for broadband PLC systems. In [8], the algorithm for bit and power allocations for an indoor narrowband PLC systems is proposed to maximize the data rate while satisfying specific spectral and bit error rate constraints at a price of a very limited complexity. In [9], optimization algorithms to improve the achievable data rate by disabling subcarriers in frequencies with low access impedance is introduced.

However, resource allocation algorithms optimizing the transmission power and the subcarrier allocation in the PLC systems have been studied as an existing optimization strategy, which is more suitable for wireless communications sometimes. Compared to wireless communications, difficulties on the optimal resource allocation by the power line (PL) channel, such as cable attenuation, various noise types and channel impedance which represents impedance behaviors or impedance characteristics at PL communication frequencies [10], are evident. Further, the channel impedance fluctuates markedly depending on the characteristics of the PL being connected to the receiver [3] and affects an amount of transmission power delivered to the receiver. Thus, to efficiently maximize EE of PLC systems, the optimization strategy should be designed with considering the characteristics of the channel impedance of the power line.

The final goal of this paper is to improve EE and provide an energy-efficient transmission strategy for EE maximization under multi-receiver orthogonal frequency division multiplexing (OFDM) downlink PLC systems. To achieve the goal, we approach through following three steps as contributions of this paper.

Step 1: We propose a dynamic load based PLC system model as a new model for EE maximization in PLC systems. The motivation to propose the new model is that an amount of received power at a load impedance of each receiver is influenced by channel characteristics of a PL channel, which

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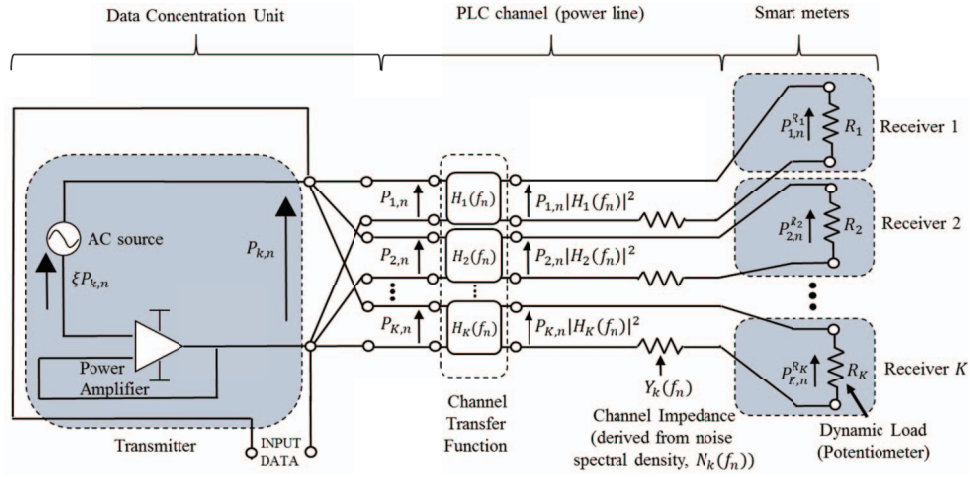


Fig. 1. The proposed multi-receiver PLC system model with dynamic load.

represents we need to estimate the optimal value of load impedance of the receivers to maximally induce the received power while considering the channel characteristics. To realize the proposed model, we need to switch the load impedance of the receiver to a potentiometer to make the impedance an optimization parameter. Further, another motivation is that the proposed model is designed to take synergistic effects of the impedance optimization and the optimal subchannel allocation in terms of efficiently maximizing EE of the PLC systems.

Step 2: We formulate an EE maximization problem including the mathematical formulation of the channel impedance varying from the channel characteristics of the PL. To reflect practical issues of the PLC systems in formulating the problem, the constraints that transmission power of a transmitter cannot be exceeded by its maximum limit and the pre-defined minimum quality of service in terms of channel capacity should be guaranteed for all receivers are involved in. Through the formulated EE maximization problem, we optimize the three arguments; load impedance, transmission power and subchannel allocation.

Step 3: We propose an optimal resource allocation algorithm as a tractable solution composed of an outer-loop algorithm to make the primal EE maximization problem convex and an inner-loop algorithm to estimate optimal resources. We studied and proposed the inner- and outer-loop algorithms based on nonlinear fractional programming and Lagrange dual method, respectively, to obtain the optimal value of the three arguments. With these three steps, we successfully acquired the optimal value of the three arguments and maximum EE of the PLC networks.

The rest of the paper is organized as follows. In Section II, we propose system models and formulate EE. In Section III, we give a solution for EE optimization and resource allocation algorithms. In Section IV, simulation results are presented. Finally, we conclude the paper in Section V.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Dynamic Load Based PLC System Model

We propose a new PLC system model considering a new argument “load impedance” to the optimization problem.

Fig. 1 shows the proposed multi-receiver OFDM downlink PLC network based on the residential automatic demand response (ADR) architecture over AMI network [3] and belongs to neighborhood area network (NAN) domain. The proposed system can be applied to low-voltage (LV) broadband PLC networks consisting of three components; data concentration unit (DCU), PLC channel and smart meter. In Fig. 1, the DCU as a transmitter in this paper is on the low-voltage side of a medium-voltage (MV)/LV transformer and composed of an AC source and power amplifier. The transmitter generating information signals is connected to the power grid and the input signal power is delivered to K receivers equally. The PLC channel consists of the channel transfer function and the channel impedance which is considered for main factor to optimize the load impedance R_k at receivers. The channel impedance derived by the background noise, $N_k(f_n)$, of the PL channel has effect on transferring the amount of the transmission power to receivers, which represents it affects EE maximization. Note that the PLC channel is independent each other. In Fig. 1, the smart meters including a PLC module are composed of K smart meters as receivers in this paper, $K \in \{1, \dots, K\}$. Each receiver receives the transmission power $P_{K,n}^R$ to its load impedance R_k which is the system impedance of the receiver k initially set to 75 [ohm] from the universal standard nominal impedance. For the effective EE maximization, we change the load impedance R_k to a potentiometer and estimate an optimum value of R_k through the proposed resource allocation strategy in this paper. The role of R_k is providing additional gain of EE by being optimized by the proposed resource allocation algorithm.

As one of solutions to dynamically control the load impedance of receivers in a practical aspect, first we need to switch the load impedance to a potentiometer. Secondly, an application of the automatic gain control (AGC) module [11] already set in the smart meter is required to practically control the value of the load as a low cost solution by using the gain control mechanism. Please note the proposed system is highly desirable in terms of energy efficiency improvements though the solution can increase the cost of the system.

B. Multi-Receiver PLC Channel Model

The impulse response of the frequency-selective power line channel, which is an inverse Fourier transfer function $H_k(f_n)$, is modeled for 1 to 30 [MHz] as follows [12], [13].

$$H_k(f_n) = A \sum_{i=1}^{\chi_i} g_i \cdot e^{-(\psi_{k,0} + \psi_{k,1} f_n^{L_k}) d_i} \cdot e^{-j2\pi f_n \left(\frac{d_i}{v}\right)}, \quad (1)$$

where $f_n = f_o + nW$ is the n -th subchannel with a start frequency of given bandwidth f_o , $W = B/N$ is the subchannel bandwidth split into a set of subchannels, $\mathcal{N} = \{1, \dots, N\}$, from the total available bandwidth B . A is a term for normalization. χ_i is the number of paths. $-1 \leq g_i \leq 1$ is a weighting factor for reflection on path i . v is $c/\sqrt{\epsilon_r}$ with the speed of light c and the dielectric constant ϵ_r . d_i is the length of the path. The attenuation parameters, $\psi_{k,0}$, $\psi_{k,1}$ and L_k , are fixed values to model a specific network and depending on types of power-line cables. Note that signal attenuation becomes larger and energy efficiency of a performance measure of the proposed system becomes smaller when the attenuation parameters are bigger. The parameters in (1) are considered random variables, and path lengths are assumed from a Poisson arrival process with intensity Λ in $[m^{-1}]$ [12]. We assume that the power line background noise exists as non-white Gaussian noise. We use a commonly accepted model for the background noise $N_k(f_n)$ which is mathematically approximated by a simple three-parameter model for 1 to 30 [MHz] [14] as follows:

$$N_k(f_n) = x + y|f_n|^z \left[V^2/\text{Hz} \right], \quad (2)$$

where x, y and z are parameters from environmental measurements. Note that the proposed systems time-periodically update the channel state information including the noise parameters by using a channel estimation scheme [15] and estimate optimum resources for EE maximization every scheduling time slot t .

C. Channel Capacity

From Fig. 1, as a multi-receiver single-input single-output system, the equivalent circuit design shows that the transmission power of the input signals, $P_{k,n}$, generated by the transmitter is delivered to all users equally. Then, the power $P_{k,n}|H_k(f_n)|^2$ is transferred by the channel transfer function, and the receiver k finally receives the transmission power $P_{k,n}^{R_k}$.

$$P_{k,n}^{R_k} = P_{k,n}|H_k(f_n)|^2 * \frac{R_k}{Y_k(f_n) + R_k}, \quad (3)$$

where $\frac{R_k}{Y_k(f_n) + R_k}$ is the power delivery rate to the impedance R_k when the channel impedance consumes a portion of the power $P_{k,n}|H_k(f_n)|^2$. Note that $Y_k(f_n) = \frac{\pi N_k(f_n)}{k_B T_0}$ [ohm] ≥ 0 [16] is the equivalent channel impedance varying from the subchannel f_n allocated by the proposed resource allocation algorithms, where $N_k(f_n)$ [V^2/Hz] is the noise spectral density, k_B [$J \cdot K^{-1}$] is the Boltzmann constant and T_0 [K] is the standard noise temperature, 290 degrees Kelvin. In fact the noise $N_k(f_n)$ is not due to true thermal noise but we assume that the noise is simply an equivalent to the electronic noise generated by typical sources of PLC noise including switching power supplies, silicon-controlled rectifiers, brush motors and dimmer

switches [17]. Therefore, the SNR at the k -th receiver for the n -th subchannel from (3) is as follows:

$$SNR_{k,n} = \frac{P_{k,n}^{R_k}}{N_k(f_n)/R_o * W}, \quad (4)$$

where $R_o = 75$ [ohm] is the initial load impedance of the receivers. Further, using the Shannon formula, the instantaneous channel capacity, C_k , for receiver k is:

$$C_k(P_{k,n}, \delta_{k,n}, R_k) = \sum_{n=1}^N \delta_{k,n} W \log_2(1 + SNR_{k,n}), \quad (5a)$$

$$C(P_{k,n}, \delta_{k,n}, R_k) = \sum_{k=1}^K C_k(P_{k,n}, \delta_{k,n}, R_k), \quad (5b)$$

where $\delta_{k,n} \in \{0, 1\}$ is a binary indicator assigning the n -th subchannel to one of the receivers, C is the network channel capacity as a summation of C_k .

D. Power Consumption

To estimate the practically expected total power consumption P , we consider the power consumption model according to [18].

$$P(P_{k,n}, \delta_{k,n}) = \underbrace{\xi \sum_{n=1}^N \sum_{k=1}^K \delta_{k,n} P_{k,n}}_{\text{Power amplifier}} + \underbrace{\sum_{k=1}^K P_{c,k}^{Tx}}_{\text{receiver}}, \quad (6)$$

where $\xi > 1$ denotes the inefficiency of the power amplifier, the static circuit power dissipation for the transmitter P_c^{Tx} and the receivers $P_{c,k}^{Rx}$ occurs in electronic components of the active transceiver, such as mixers, filters and digital-to-analog converters, and is independent in the actual transmission power $P_{k,n}$.

E. Network Energy Efficiency

The resource allocation problem for EE maximization of the proposed PLC network is formulated as:

$$(P1) \quad \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \eta = \frac{C(P_{k,n}, \delta_{k,n}, R_k)}{P(P_{k,n}, \delta_{k,n})}, \quad (7)$$

subject to

$$(C1) \quad C(P_{k,n}, \delta_{k,n}, R_k) \geq C^{QoS}, \quad (8)$$

$$(C2) \quad \sum_{n=1}^N \sum_{k=1}^K \delta_{k,n} P_{k,n} \leq P_{max}, P_{k,n} \geq 0. \quad (9)$$

$$(C3) \quad \xi \sum_{n=1}^N \sum_{k=1}^K \delta_{k,n} P_{k,n} + P_c^{Tx} \leq P_{pg}. \quad (10)$$

$$(C4) \quad \sum_{k=1}^K \delta_{k,n} \leq 1, \delta_{k,n} \geq 0, \forall n \in \mathcal{N}. \quad (11)$$

$$(C5) \quad \delta_{k,n} = \{0, 1\}, \quad \forall n \in \mathcal{N}, \forall k \in \mathcal{K}. \quad (12)$$

The energy efficiency η is constructed as a summation of the channel capacity per a summation of consumed power for the transmitter and all receivers, and the unit of η is [bits/Joule]. The constraint C1 implies that the minimum network channel

capacity C^{QoS} should be guaranteed. The constraint C2 represents that the total transmission power cannot be exceeded by the maximum transmission power P_{max} [19]. Constraint C3 limits the total power consumption of the transmitter to not exceed P_{pg} which is the maximum amount of the electricity that can be delivered from the power grid. Constraint C4 indicates that each subchannel can be allocated to only one receiver to prevent interference among receivers. C5 is a combinatorial constraint on the subcarrier assignment.

III. ENERGY EFFICIENCY MAXIMIZATION

In this section, we solve the defined problem P1 and propose an EE optimization algorithm estimating the optimum transmission power, the optimum subcarrier allocation and the optimum load impedance. First of all, since solving the non-convex objective function P1 is extremely challenging and the optimum values can be found using a brute-force approach only, we transform the P1 to the convex function using nonlinear fractional programming [20] in Section III-A. In Section III-B, we introduce Lagrangian dual decomposition as a greedy-type iterative solution to estimate optimum resources, $\{P_{k,n}, \delta_{k,n}, R_k\}$, guaranteeing maximum EE. With these two optimization methods, the optimal value of the three arguments, $\{P_{k,n}, \delta_{k,n}, R_k\}$, and maximum energy efficiency can be successfully acquired.

A. Outer-Loop Algorithm: Transformation of the Primal Objective Function

In this subsection, we propose an effective and tractable solution to solve the non-convex objective function P1 based on nonlinear fractional programming [20]. The objective function P1 is equivalent to $C(P_{k,n}, \delta_{k,n}, R_k) - \eta^* P(P_{k,n}, \delta_{k,n})$ by exploiting properties of nonlinear fractional programming, where η^* is an optimum solution of P1 when the argument set $\{P_{k,n}, \delta_{k,n}, R_k\}$ is equal to the optimal argument set $\{P_{k,n}^*, \delta_{k,n}^*, R_k^*\}$. To mathematically define that the problem P1 has the equivalent and convex function, we define F as a set of feasible solutions and maximum EE as η^* in the optimization problem P1. The theorem for the transformation is as follows.

Theorem 1:

$$\eta^* = \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \left\{ \frac{C(P_{k,n}, \delta_{k,n}, R_k)}{P(P_{k,n}, \delta_{k,n})} \mid \forall \{P_{k,n}, \delta_{k,n}, R_k\} \in F \right\}$$

$$= \frac{C(P_{k,n}^*, \delta_{k,n}^*, R_k^*)}{P(P_{k,n}^*, \delta_{k,n}^*)}$$

if and only if, $\max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \{C(P_{k,n}, \delta_{k,n}, R_k) - \eta^* P(P_{k,n}, \delta_{k,n})\} = C(P_{k,n}^*, \delta_{k,n}^*, R_k^*) - \eta^* P(P_{k,n}^*, \delta_{k,n}^*) = 0$, for $C(P_{k,n}, \delta_{k,n}, R_k) \geq 0$ and $P(P_{k,n}, \delta_{k,n}) > 0$.

Theorem 1 represents that the objective function P1 in fractional form can equivalently be transformed in subtractive form. For a proof of Theorem 1, the transformed function $F(\eta)$ can be defined as follows:

$$F(\eta) = \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \{C(P_{k,n}, \delta_{k,n}, R_k) - \eta P(P_{k,n}, \delta_{k,n})\}. \quad (13)$$

TABLE I
OUTER-LOOP ALGORITHM FOR EFFECTIVE OPTIMIZATION

1: Initialize $\eta = \frac{C(P_{k,n}, \delta_{k,n}, R_k)}{P(P_{k,n}, \delta_{k,n})} = 0, \tau \approx +0$ (positive zero).
2: Update $P_{k,n}^*, \delta_{k,n}^*$ and R_k^* by solving the inner-loop algorithm.
3: If $C(P_{k,n}^*, \delta_{k,n}^*, R_k^*) - \eta P(P_{k,n}^*, \delta_{k,n}^*) < \tau$
4: Return Optimal resources $\{P_{k,n}^*, \delta_{k,n}^*, R_k^*\}$ and optimal EE, $\eta^* = \frac{C(P_{k,n}^*, \delta_{k,n}^*, R_k^*)}{P(P_{k,n}^*, \delta_{k,n}^*)}$.
5: else Update $\eta = \frac{C(P_{k,n}^*, \delta_{k,n}^*, R_k^*)}{P(P_{k,n}^*, \delta_{k,n}^*)}$ and Go to line 2.

By following approaches shown in [20], we can proof Theorem 1.

Proof: Convergence and maximization. ■

Lemma 1: $F(\eta)$ is exactly monotonic decreasing in η ; i.e., $F(\eta') > F(\eta'')$ if $\eta' < \eta''$.

Proof: Let η'' maximize $F(\eta'')$, then

$$\begin{aligned} F(\eta'') &= \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} C(P_{k,n}, \delta_{k,n}, R_k) - \eta'' P(P_{k,n}, \delta_{k,n}) \\ &= C(P_{k,n}'', \delta_{k,n}'', R_k'') - \eta'' P(P_{k,n}'', \delta_{k,n}'') \\ &< C(P_{k,n}'', \delta_{k,n}'', R_k'') - \eta' P(P_{k,n}'', \delta_{k,n}'') \\ &\leq \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \{C(P_{k,n}, \delta_{k,n}, R_k) - \eta' P(P_{k,n}, \delta_{k,n})\} \\ &= F(\eta') \end{aligned}$$

where $C(P_{k,n}'', \delta_{k,n}'', R_k'') - \eta'' P(P_{k,n}'', \delta_{k,n}'') < C(P_{k,n}'', \delta_{k,n}'', R_k'') - \eta' P(P_{k,n}'', \delta_{k,n}'')$ is reasonable because η' is smaller than η'' , as stated in Lemma 1. ■

Lemma 2: Let any set $\{P_{k,n}', \delta_{k,n}', R_k'\}$ and the set satisfies $\eta' = \frac{C(P_{k,n}', \delta_{k,n}', R_k')}{P(P_{k,n}', \delta_{k,n}')}$, then $F(\eta') \geq 0$.

Proof:

$$\begin{aligned} F(\eta') &= \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} C(P_{k,n}, \delta_{k,n}, R_k) - \eta' P(P_{k,n}, \delta_{k,n}) \\ &\geq C(P_{k,n}', \delta_{k,n}', R_k') - \eta' P(P_{k,n}', \delta_{k,n}') = 0. \end{aligned}$$

As shown in Lemma 1 and Lemma 2, it is natural that $F(\eta)$ converges to 0 because $F(\eta)$ is a monotonic decreasing function as shown in Lemma 1 and $F(\eta) \geq 0$ as shown in Lemma 2. Further, what $F(\eta)$ converges to 0 represents that η goes to maximum value. Thus, we define P2 as an equivalent problem P1 of the transformed objective function $F(\eta)$ as follows:

$$(P2) \quad \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \{C(P_{k,n}, \delta_{k,n}, R_k) - \eta P(P_{k,n}, \delta_{k,n})\},$$

subject to (C1), (C2), (C3), (C4), (C5), (14)

where $C(P_{k,n}, \delta_{k,n}, R_k) - \eta P(P_{k,n}, \delta_{k,n}) \geq 0$ is given for any value η generated by the outer loop algorithm which is shown in Table I. We note that the problem P2 is a convex and combinatorial optimization problem weighted by η for EE maximization. However, the problem P2 still has a combinatorial problem to optimally allocate the subchannels, which can be viewed as searching for the best case for EE maximization. To make the combinatorial constraint C5 tractable, we

relax constraint C5 according to the approach in [21]. Instead of Boolean logic of C5, now we let $\delta_{k,n}$ be a real number between 0 and 1 by using continuous relaxation. Thus, we rewrite the problem P2:

$$(P3) \quad \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \{C(P_{k,n}, \delta_{k,n}, R_k) - \eta P(P_{k,n}, \delta_{k,n})\}, \quad (15)$$

subject to (C1), (C2), (C3), (C4),

$$(C5) \quad 0 \leq \delta_{k,n} \leq 1, \forall n, k. \quad (16)$$

We can easily confirm that the transformed problem P3 is a jointly convex optimization problem as it respects all optimization variables under [22]. We use the optimization problem P3 as an equivalent objective function P1 in the rest of this paper. Further, we propose the outer-loop algorithm to effectively solve the optimization problem P1 based on Theorem 1, and the pseudo-code of the outer-algorithm is listed in Table I. According to the proposed algorithm, the convergence to maximum EE is guaranteed if the inner-loop problem P3 in each iteration is solved in Table I, Line 2. When the outer-loop iterations go on, the η increases, and it converges to η^* if $F(\eta) < \tau$ in Table I, Line 3 and 4. Note that the threshold parameter τ for the convergence of the function $F(\eta)$ should be set to around positive zero, $\tau \approx +0$. In conclusion, $F(\eta)$ will finally approach zero and we can obtain maximum EE. ■

B. Inner-Loop Algorithm: Resource Allocations

In this subsection, we solve the convex optimization problem P3 as the inner problem in Table I, Line 2 by using the Lagrange dual method. Also, we propose the inner-loop algorithm estimating the optimal arguments, $P_{k,n}^*$, $\delta_{k,n}^*$ and R_k^* , respectively. The Lagrange dual function of P3 is derived as:

$$\begin{aligned} \mathcal{L}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n) &= \sum_{n=1}^N \sum_{k=1}^K \mathcal{L}_{k,n}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n) - \alpha C^{QoS} + \beta P_{max} \\ &+ \gamma (P_{pg} - P_c^{Tx}) + \sum_{n=1}^N \theta_n - \eta \left(P_c^{Tx} + \sum_{k=1}^K P_{c,k}^{Rx} \right), \end{aligned} \quad (17a)$$

with

$$\begin{aligned} \mathcal{L}_{k,n}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n) &= -(\eta\xi + \beta + \gamma\xi)\delta_{k,n}P_{k,n} \\ &+ (1 + \alpha)\delta_{k,n}W\log_2(1 + SNR_{k,n}) - \delta_{k,n}\theta_n, \end{aligned} \quad (17b)$$

where α , β , γ and θ_n are Lagrange multipliers. When we derived the optimal solution by Lagrange dual method, the Karush–Kuhn–Tucker (KKT) conditions absorb the boundary constraint C5. Therefore, the dual problem of P3 is as follows:

$$\min_{\{\alpha, \beta, \gamma, \theta_n\}} \max_{\{P_{k,n}, \delta_{k,n}, R_k\}} \mathcal{L}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n). \quad (18)$$

In the following, we can solve inner loop algorithm with the dual problem (18) by decomposing it into two layers (Layer 1: sub problem, Layer 2: super problem). According to the decomposition of Equation (18), we solve Layer 1: sub problem by using KKT conditions. Let the optimal solution of

the sub problem be $P_{k,n}^*$, $\delta_{k,n}^*$ and R_k^* then the KKT conditions reveal that:

$$\begin{aligned} \frac{\partial \mathcal{L}_{k,n}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n)}{\partial P_{k,n}} &= \frac{W\delta_{k,n}(1 + \alpha)|H_k(f_n)|^2 R_k R_o}{(WN_k(f_n)(Y_k(f_n) + R_k) + |H_k(f_n)|^2 R_k P_{k,n}) \ln 2} \\ &- (\beta + \gamma\xi + \eta\xi)\delta_{k,n} \begin{cases} = 0, & P_{k,n} > 0 \\ < 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{\partial \mathcal{L}_{k,n}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n)}{\partial \delta_{k,n}} &= A_{k,n} - \theta_n \begin{cases} \geq 0, & 0 < \delta_{k,n} \leq 1 \\ < 0, & \delta_{k,n} = 0, \end{cases} \end{aligned} \quad (20)$$

$$\theta_n \left(\sum_{k=1}^K \delta_{k,n} - 1 \right) = 0, \quad (21)$$

$$\begin{aligned} \frac{\partial \mathcal{L}_{k,n}(P_{k,n}, \delta_{k,n}, R_k, \alpha, \beta, \gamma, \theta_n)}{\partial R_k} &= \frac{\delta_{k,n}W(1 + \alpha) \left(\frac{DP_{k,n}}{EY_k(f_n) + ER_k} - \frac{DP_{k,n}ER_k}{(EY_k(f_n) + ER_k)^2} \right)}{\ln 2 * \left(\frac{DP_{k,n}R_k}{EY_k(f_n) + ER_k} + 1 \right)} \\ &\begin{cases} = 0, & R_k > 0 \\ < 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (22)$$

where $A_{k,n} = W(1 + \alpha)\log_2(1 + SNR_{k,n}) - (\gamma\xi + \eta\xi + \beta)P_{k,n}$, $D = |H_k(f_n)|^2 R_o$ and $E = WN_k(f_n)$. By the KKT conditions, the optimal transmission power $P_{k,n}^*$ is calculated by jointly solving the equalities from (19) to (22).

$$P_{k,n}^* = \delta_{k,n}^* \left[\frac{W(1 + \alpha)R_o}{(\beta + \gamma\xi + \eta\xi)\ln 2} - \frac{E(Y_k(f_n) + R_k)}{DR_k} \right]^+. \quad (23)$$

As a form of multilevel water filling, the optimal transmission power allocation is carried out in (23). From (20), $A_{k,n} \geq 0$ is marginal benefits [23] in allocating the n -th subchannel to receiver k . The physical meaning of $A_{k,n}$ is the rate of change of the channel capacity with regard to $\delta_{k,n}$. If $A_{k,n}$ of the subchannel is lower than zero, we can assume that subchannel has the poor channel condition and the channel capacity is being degenerated. According to the physical meaning of $A_{k,n}$, an optimal channel allocation strategy for n -th subcarrier is given by:

$$\delta_{k,n}^* = \begin{cases} 1, & \text{if } A_{k,n} = \max_j A_{j,n} \\ 0, & \text{otherwise,} \end{cases} \quad (24)$$

where j represents all active receivers, and only receivers who have the largest marginal benefits on the network are selected by the subchannel allocator $\delta_{k,n}^*$. Then, with optimally allocated subchannel number n^* for receiver k , the channel transfer function $H_k(f_n)$ are updated to $H_k(f_{n^*})$. Using the updated channel transfer function, R_k^* calculated by the KKT conditions is also updated as follows:

$$R_k^* = \frac{E * Y_k(f_{n^*}) - 2^{\frac{(F * P_{k,n} + \theta_n)}{W(1 + \alpha)}} * E * Y_k(f_{n^*})}{2^{\frac{(F * P_{k,n} + \theta_n)}{W(1 + \alpha)}} * E - D * P_{k,n} - E}, \quad (25)$$

where $F = \beta + \gamma\xi + \eta\xi$.

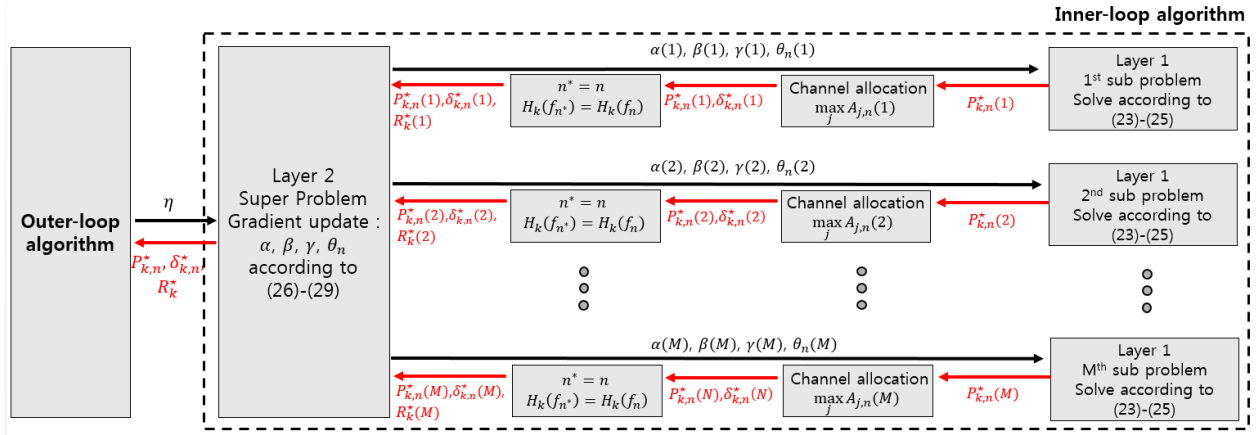


Fig. 2. The diagram of the proposed iterative algorithms for EE optimization.

After solving the sub problem (Layer 1), the super problem (Layer 2) adjusts the Lagrange multipliers. α , β , γ and θ_n are updated by gradient methods in a distributed manner, which are as follows:

$$\alpha(T_{in}+1) = \left[\alpha(T_{in}) - \mu_1 \left(C(P_{k,n}^*, \delta_{k,n}^*, R_k^*) - C^{QoS} \right) \right]^+, \quad (26)$$

$$\beta(T_{in}+1) = \left[\beta(T_{in}) - \mu_2 \left(P_{max} - \sum_{n=1}^N \sum_{k=1}^K \delta_{k,n}^* P_{k,n}^* \right) \right]^+, \quad (27)$$

$$\gamma(T_{in}+1) = \left[\gamma(T_{in}) - \mu_3 (P_{pg} - P(P_{k,n}^*, \delta_{k,n}^*)) \right]^+, \quad (28)$$

$$\theta_n(T_{in}+1) = \left[\theta_n(T_{in}) - \mu_4 (1 - \delta_{k,n}^*) \right]^+, \quad (29)$$

where $[Z]^+ = \max[Z, 0]$, T_{in} is the number of inner-loop iterations in which the transmitter updates α , β , γ and θ_n . The constant coefficients μ_1 , μ_2 , μ_3 and μ_4 are positive iteration steps.

From the above optimal resource allocation strategy, we propose the pseudo-code of the iterative inner-loop algorithm in Table II. The inner-loop algorithm is performed in Table I, Line 2, and overall operating figure by the proposed outer- and inner-algorithm is described in Fig. 2.

IV. SIMULATION RESULTS

In this section, we provide numerical results to show the effectiveness of the proposed algorithm. The system is set up as follows. The transmitter is serving six receivers ($K=6$) as a neighborhood area network size having not less than ten receivers. Further, we set the different attenuation parameters to each receiver to assign dynamic channel environments in a typical range based on [12]. The attenuation parameters are an important parameter to monitor how the optimization works when the various channel environments are given. The attenuation parameters for the k -th receiver are set to $(\psi_{k,0}, \psi_{k,1}) = (10^{-5}, 10^{-9})$, $(10^{-4.3}, 10^{-9.2})$, $(10^{-5.6}, 10^{-8.5})$, $(10^{-5.4}, 10^{-9.8})$, $(10^{-3.9}, 10^{-9.4})$, $(10^{-5.2}, 10^{-10.1})$, respectively. The noise parameters is set to $(x, y, z) = (-143, 47.2, -0.536)$ in a typical range based on [13]. $P_{k,n}(1) = 2.25$ [mW]. $R_k(1) = R_o = 75$ [ohm]. $\xi = 4$.

TABLE II
INNER-LOOP ALGORITHM FOR RESOURCE ALLOCATION

- 1: **Input** $C^{QoS}, P_{max}, P_{c,k}^{Rx}, P_c^{Tx}, P_{pg}, \xi, B, N_k(f_n), A, N_i, \psi_{k,0}, \psi_{k,1}, \varepsilon_r, g_i, d_i, f_o, L_k, T_o, k_B, x, y, z$.
- 2: **Initialize** Active UE set $k = \{1, 2, \dots, K\}$, $P_{k,n}(1) > 0$, $\delta_{k,n}(1) > 0$, $R_k(1) > 0$, $\mu_1 > 0$, $\mu_2 > 0$, $\mu_3 > 0$, $\mu_4 > 0$, $\alpha(1) = \beta(1) = \gamma(1) = \delta_n(1) > 0$.
- 3: **While** α_k, β, γ and θ_n are not converged **do**
- 4: **for** $k \in \mathcal{K}$ **do**
- 5: **for** $n \in \mathcal{N}$ **do**
- 6: Update $P_{k,n}^* = P_{k,n}(T_{in} + 1)$ according to (23).
- 7: Update $\delta_{k,n}^* = \delta_{k,n}(T_{in} + 1)$ according to (24).
- 8: **end for**
- 9: Update $R_k^* = R_k(T_{in} + 1)$ according to (25).
- 10: **end for**
- 11: **for** $n \in \mathcal{N}$ **do**
- 12: Update the subgradient of $\theta_n(T_{in} + 1)$ with (29).
- 13: **end for**
- 14: Update the subgradient of $\alpha(T_{in} + 1)$ with (26).
- 15: Update the subgradient of $\beta(T_{in} + 1)$ with (27).
- 16: Update the subgradient of $\gamma(T_{in} + 1)$ with (28).
- 17: Let $T_{in} = T_{in} + 1$.
- 18: **end while**

$P_{c,k}^{Rx} = 24$ [dBm]. $P_c^{Tx} = 33$ [dBm]. $P_{pg} = 50$ [dBm]. $L_k = 1$. $C^{QoS} = 60$ [Mbps]. The bandwidth B is from 1 to 30 [MHz] with the number of subcarriers $N = 256$, and subchannel bandwidth $W = 110$ [kHz]. In the following results, we set the maximum number of inner-loop iterations to six.

Fig. 3 demonstrates the evolution of the proposed iterative algorithms for different baseline schemes with the maximum transmission power of $P_{max} = 36$ [dBm] at the transmitter. "Iteration" is the number of iterations in Table I. The result was averaged over 1,000 independent simulations, each of which involved different channel and noise parameters. From Fig. 3, it is observed the maximum EE can be achieved within six iterations on average. Fig. 3 also covers the EE of baseline resource allocation schemes. We can obtain the baseline schemes as the optimal resource allocation strategy by fixing one of the arguments to a constant. The fixed value of R_k

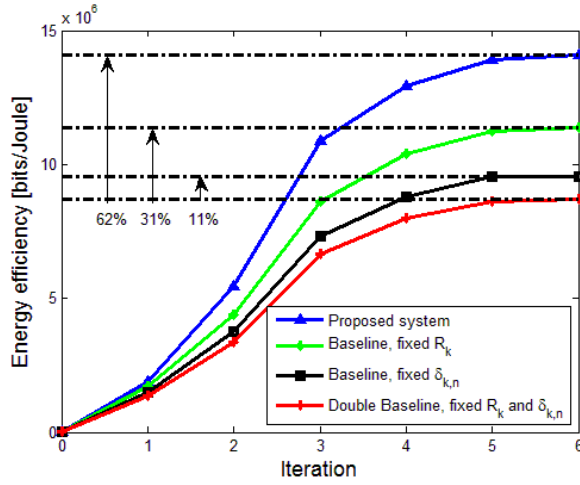


Fig. 3. EE [bits/Joule] versus iterations with $K = 6$ receivers. ($P_{max} = 36$ [dBm]).

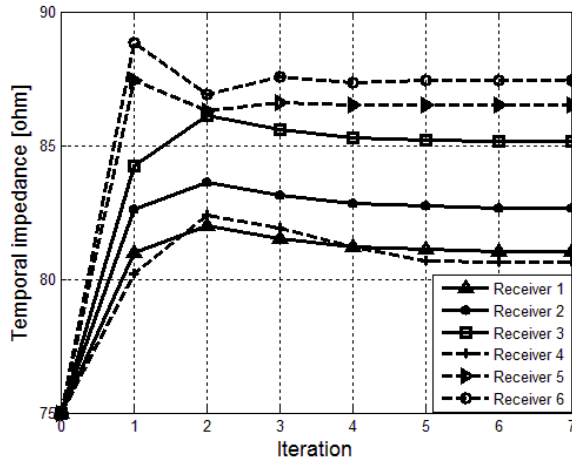


Fig. 4. Temporal impedance [ohm] versus iterations.

is 75 [ohm] and equal bandwidth is distributed to receiver k . The proposed system provides a significant performance gain over the baseline schemes with respect to energy efficiency. Furthermore, it is observed that the proposed system creates a synergy effect on energy efficiency when the impedance optimization is adopted by the system. This is because the channel impedance is involved in subcarrier allocations.

Fig. 4 shows the evolution of the temporal impedance for receiver k according to each iteration. The optimization is in progress to obtain the maximum EE from the initial impedance of 75 [ohm] with the constraint C1 for the minimum network capacity C^{QoS} satisfied. The maximum EE can be achieved by obtaining maximum $SNR_{k,n}$ calculated by the proposed resource allocation algorithms as optimal ratio of received signal power to noise power. From the optimal impedances in Fig. 4, we can provide the maximized $SNR_{k,n}$ to the receiver k .

Fig. 5 depicts the energy efficiency versus the number of users. Different baseline schemes, network data rate requirements C^{QoS} , and six iterations were considered in this simulation. When the number of receivers increases, energy

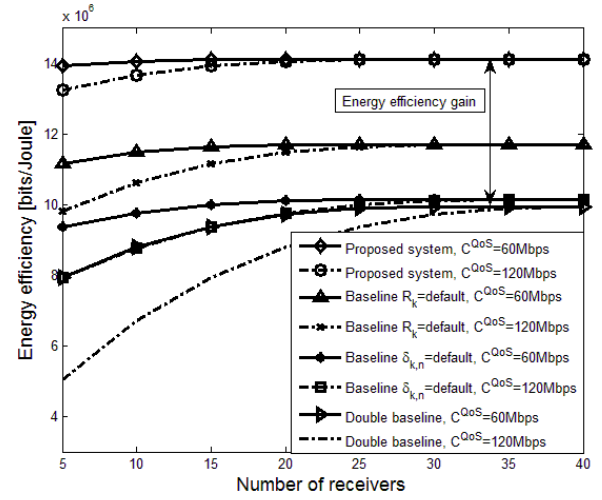


Fig. 5. EE versus the number of receivers with single and double baseline schemes.

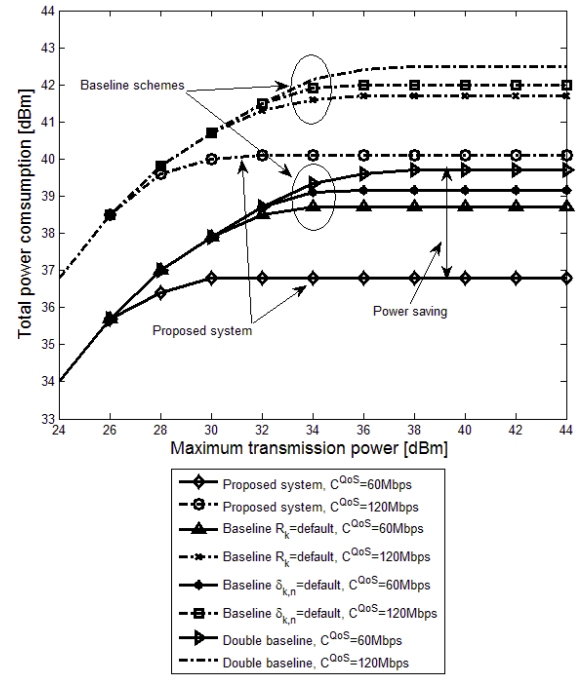


Fig. 6. Maximum transmission power [dBm] versus total power consumption [dBm] with single and double baseline schemes.

efficiency also increases because the proposed resource allocation algorithms can obtain multi-user diversity (MUD) by introducing an extra power gain to the system. Therefore, when the number of receivers is sufficiently large, the EE eventually approaches its limit. Further, the proposed system is advantageous to achieve maximum energy efficiency despite a small number of receivers since the energy efficiency of the proposed system attains the EE limit rapidly.

Fig. 6 shows the total power consumption $P(P_{k,n}, \delta_{k,n})$ for the proposed system and the Baseline schemes when the maximum transmission power P_{max} increases. Particularly in the regime of high transmission power, the result shows that the baseline schemes consumes more power than the

proposed system. Furthermore, the increasing transmission power results in greater power savings in achieving the target data rate. Note that the power savings are much larger when the subchannel allocation strategy $\delta_{k,n}$ and impedance adaptation are simultaneously considered. This has the same cause as the synergy effects in the description of Fig. 3.

V. CONCLUSION

In this paper, for energy-efficient PLC systems, we have proposed the dynamic load based PLC system model as a new model for EE maximization and formulated the energy-efficient transmission strategy optimizing the transmission power, subchannel allocation and load impedance. As a combinatorial and non-convex maximization problem, we adopted nonlinear fractional programming and dual method to obtain the tractable solution. Simulation results showed that the proposed system is energy-efficient, and energy efficiency is greatly increased by the synergy effects derived by the impedance optimization and the subchannel allocation strategy.

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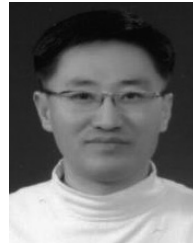


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